



Plastic zone characteristics and rapid stability evaluation indexes of circular tunnels subjected to non-hydrostatic stress: A novel complex variable solution incorporating 3D rock mass strength

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ABSTRACT

The morphological characteristics and distribution range of the plastic zone around tunnels, influenced by the true three-dimensional high geostress environment in deep engineering, constitute fundamental reference data for both tunnel design and the control of surrounding rock stability. Based on the GZZ three-dimensional rock mass strength criteria and the complex variable method, a novel semi-analytical method for solving the plastic zone of a circular tunnel subjected to non-hydrostatic stress is presented. The conformal mapping is employed to intricately transform the complex boundaries of plastic zones characterized by diverse shapes onto the unit circle boundary within the image plane. This framework proves invaluable for the precise analytical process of the complex plastic zone morphology and the plastic influence range within the surrounding rock under complex stress conditions. In contrast to traditional analytical methods, the efficacy and versatility of our proposed approach are substantiated. Notably, it demonstrates unique advantages, particularly in the context of non-hydrostatic stress fields and varied rock mass parameters, with respect to predicting both surrounding rock stability and plastic zone morphology. Through the analysis of numerous cases, the study reveals the systematic patterns of influence that in-situ stress and surrounding rock characteristic parameters have on the shape and maximum depth of the plastic zone. The results indicate that (1) there are notable disparities in the morphology and extent of the plastic zone between the GZZ and GHB criteria; (2) the transition of the plastic zone in surrounding rock from U-shaped to V-shaped is significantly influenced by the geostress-to-strength ratio K_σ ; (3) there is a notable linear correlation observed between the maximum plastic zone thickness $r_{\max}$ and both K_σ and the stress concentration factor K_{SCF} , importantly, this correlation is independent of far-field stress and the brittleness characteristics of the surrounding rock. These findings provide a novel discriminative method for the rapid analysis of surrounding rock stability in deep circular tunnels.

1. Introduction

The stability analysis of underground deep-buried tunnels under high and complex stress condition is imperative for ensuring the secure implementation of tunnel construction practices. The investigation on the deformation and failure behavior of the surrounding rock requires a precise description of the post-peak mechanical responses exhibited by the surrounding rock mass. As is widely acknowledged, the constitutive laws of rocks are commonly categorized into three classical simplified models: the ideal elastic-plastic model (EPP), the strain softening model (ESS), and the elastic-brittle-plastic model (EBP). The applicability of

these models is closely related to the intrinsic properties of rocks, the stress conditions the rocks are subjected to, and the degree of joint development within the rock mass (Alejano et al., 2010; Cheon et al., 2011; Xie et al., 2021). The brittleness of rock masses with a higher GSI value, indicating a more intact structure, is more pronounced compared to soft fractured rock masses with a lower GSI value. Therefore, it is recommended to employ the EBP model for rock masses of higher quality, while the ESS or EPP models are recommended for rock masses exhibiting medium or poor quality, respectively. Due to the computational complexity and convergence challenges caused by using the ESS model, Su et al. (2023) proposed utilizing the EPP model for an

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equivalent simulation of the strain softening model.

The stability of surrounding rock after tunnel excavation is related to factors such as geostress, rock strength, and joint development. Many scholars have dedicated efforts to identify various influencing factors on surrounding rock stability and conduct sensitivity analyses on the influencing parameters. Analytical or semi-analytical methods exhibit a pronounced advantage in rapid parameter analysis, particularly in contrast to methodologies like model tests, in-situ testing, or numerical simulations, which are constrained by equipment and time consumption (Zhang et al., 2022). Many elastic-plastic analyses (Alejano et al., 2009; Ghorbani and Hasanzadehshooiili, 2019; Lu et al., 2018; Xu and Xia, 2021) for circular tunnels have been conducted based on the simplified EPP model. However, when considering the brittle failure of rock masses, most existing analytical or semi-analytical methods (Chen and Tonon, 2011; Wang et al., 2012; Zareifard and Fahimifar, 2015; Ghorbani and Hasanzadehshooiili, 2019; Singh et al., 2019) mainly involve analysis under axisymmetric stress conditions. These methods are difficult to account for the three-dimensional complex stress environment in actual tunnels.

More importantly, numerous studies have shown that tunnels excavated in brittle hard rock typically generate U-shaped or V-shaped plastic zones in the local vicinity near the tunnel walls. Extensive plastic zones surrounding the excavation boundaries, as observed in soft rock tunnels, occur only when the in-situ stresses are sufficiently high. According to field measurements obtained during mine-by experiments at the Atomic Energy of Canada Limited Underground Research Laboratory (Martin et al., 1999; Martin and Christiansson, 2009), the initiation, expansion, and development processes of the plastic zone and damage zone in a granite circular tunnel after excavation were observed. Subsequently, many scholars (Su et al., 2017; Si and Gong, 2020; Hu et al., 2022; Li et al., 2022; Feng et al., 2023) have simulated the failure processes of tunnels under high stress conditions indoors by developing experimental equipment or improving experimental methods, and have utilized high-speed cameras to comprehensively record phenomena such as rock spalling at tunnel walls or even dynamic characteristics of brittle failures. Typically observed in hard rocks like granite, quartzite, and marble, these phenomena may also manifest in softer rocks like schist and sandstone under appropriate stress conditions. In theory, the formation of shear zones, or phenomena like slabbing or spalling roughly parallel to the excavation face, observed in most cases, is motivated by the stress or deformation concentration within narrow regions of the rock mass. In tunnel engineering design, the maximum depth of the rock mass plastic zone or failure zone is a crucial parameter of concern for designers. The plastic zone and the failure zone are fundamentally distinct, there is no doubt that the position of the failure zone is located within the plastic zone. Extensive empirical field data, however, demonstrated that the rock mass failure zone exhibits a similar shape to the plastic zone, and the ratio between the depth of the failure zone and the depth of the plastic zone falls within a certain range (Ma et al., 2023). Currently, there are some empirical or numerical back analysis methods for predicting the maximum depth of the failure zone in brittle rock masses (Martin and Christiansson, 2009; Perras and Diederichs, 2016; Walton et al., 2014). However, no analytical or semi-analytical methods have been proposed to parametrically study the variation in the maximum plastic zone depth induced by brittle failure of the surrounding rock.

In recent years, the successful application of complex variable method in the mechanical analysis of surrounding rock in deep tunnels has yielded numerous analytical solutions (Chen et al., 2022; Cheng et al., 2021; Lu et al., 2018; Zhang et al., 2022). The existing complex variable solutions (Lu et al., 2018; Zhuang and Yu, 2019) for elastic-plastic analysis of surrounding rocks mostly ignore the three-dimensional strength characteristics of deep rock masses or require limitations on geostress conditions, resulting in an insufficient understanding of the morphological characteristics and evolution laws of plastic zones under complex stress environments in deep areas. For a

precise analysis of the stability and elastoplastic stress states of surrounding rock under complex three-dimensional stress conditions, the application of rock strength theory (yield criterion) is of paramount importance. The generalized Zhang-Zhu strength criterion (GZZ) serves well in capturing the three-dimensional strength characteristics of rock masses at the engineering scale (Cai et al., 2021), and its strength parameters inherit the Hoek-Brown (HB) theory system. At present, a methodology based on the digitization of tunnel face geometry information has been developed for in-situ determination of the Geological Strength Index (GSI), providing significant advantages in three-dimensional forward analysis and dynamic design in rock engineering (Cai et al., 2022c, 2023; Chen et al., 2016; Zhang et al., 2020).

In response to the abovementioned challenges, this study focuses on circular tunnels subjected to a three-dimensional high-stress condition. The GZZ strength criterion is employed to characterize and incorporate the three-dimensional strength properties of the rock mass. Based on the complex variable method, a semi-analytical method which can accommodate both the EPP and EBP models is proposed for elastic-plastic analysis of surrounding rock. The method proposed in this paper ingeniously treats the plastic stress component along the boundary of the plastic zone as an external load acting on the inner boundary of the elastic zone of the surrounding rock, and then provides the governing equations for determining the analytical function based on the boundary conditions. Remarkably, this method does not impose special requirements on whether the plastic zone can completely enclose the tunnel wall, making it a unified approach suitable for the stability analysis of surrounding rocks around deep circular tunnels. The investigation subsequently comprehensively considered factors such as the magnitude of in-situ stress, intact rock strength, and the development of jointing within the surrounding rock. And a comparative analysis was conducted through examples, revealing the evolutionary patterns of plastic zone characteristics and stress distribution within the rock masses exhibiting both ideal elastoplastic and brittle properties. The study provides a detailed elucidation of the impact of brittle failure on the stability of the surrounding rocks.

2. Problem description

2.1. Calculation model

The simplification of the surrounding rock response to the excavation of deeply buried circular tunnels as an axisymmetric problem, treated as a “thick-walled cylinder” model, has been a prevalent topic in previous discussions. Although this model significantly reduces the difficulty of solving the problem, the resulting distribution of the plastic zone is circular, and the obtained stress and displacement fields are axisymmetric. This deviates significantly from the actual stress environment in which tunnels exist.

This paper considers the tunnel subjected to unequal in-plane horizontal stress σ_x^∞ , vertical stress σ_y^∞ , and out-of-plane stress σ_z along the tunnel axis, as illustrated in Fig. 1. For soft rock tunnels, the depth of the plastic zone is relatively large, typically forming a complete plastic zone surrounding the entire tunnel boundary. In contrast, for hard rock tunnels, the depth of the plastic zone is smaller, usually resulting in localized plastic zones around the tunnel. In addition, there are many factors that affect the morphology of the plastic zone. For example, the plastic zone tends to completely enclose the circular tunnel when the difference between horizontal and vertical ground stresses is small. The evolution of plastic zone morphology will be discussed in more detail in the examples. It is worth noting that the general method proposed in this article does not make special distinctions between the two cases in Fig. 1. Let the radius of the circular tunnel be denoted as R_0 . The boundary of the elastic zone in the surrounding rock is denoted as L_e , and the boundary of the plastic zone is denoted as L_p . That is, the inner and outer sides of the elastic-plastic boundary are L_p and L_e , respectively.

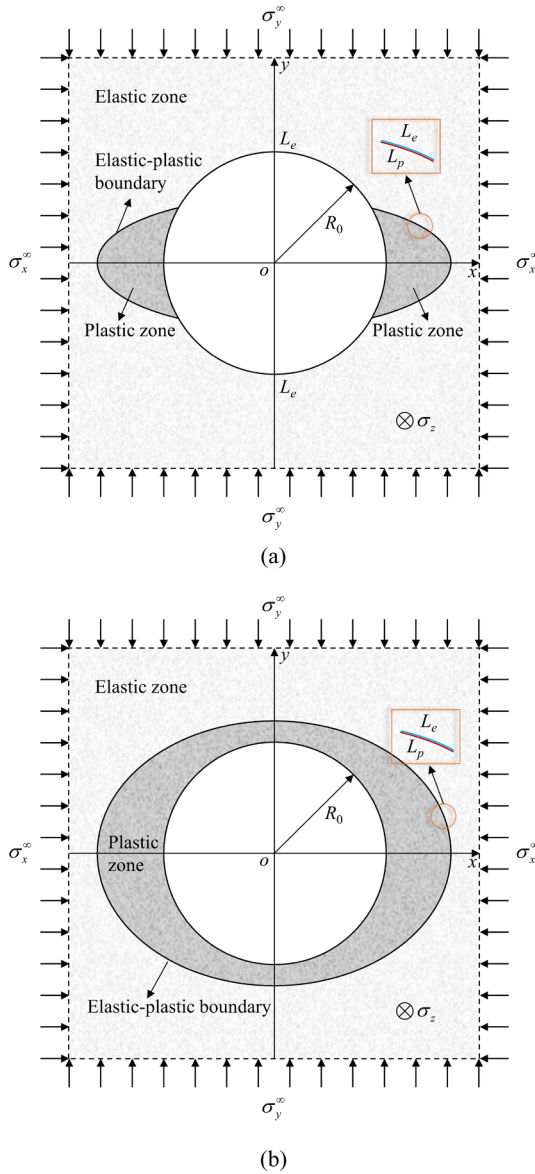


Fig. 1. Schematic diagram of the plastic zone pattern around a circular tunnel under three-dimensional stress conditions: (a) Local plastic zone; (b) Complete plastic zone.

It is pertinent to underscore that despite our systematic categorization and discussion of the distinct morphological characteristics of the two types of plastic zones mentioned above, the specific geological conditions governing whether the plastic zone can entirely envelop the tunnel wall are indeterminate when addressing a particular tunneling project. This paper utilizes general formulae when deriving the governing equations for determining the plastic zones. No distinction is made in the morphological characteristics of the plastic zones. Consequently, a unified semi-analytical solution is obtained for calculating the plastic zones of the surrounding rocks around a circular tunnel. Furthermore, our method is established under the following assumptions:

- (1) The rock mass is assumed to be a homogeneous and isotropic material, with its post-peak mechanical behavior described by described by both the elastic-perfect-plastic model and the elastic-brittle-plastic model, generally corresponding to engineering soft and hard rocks, respectively.

- (2) The deformations within the surrounding rock are significantly smaller than the tunnel dimensions. Throughout the entire elastic-plastic deformation phase, the small deformation assumption is satisfied.
- (3) The tunnel is assumed to be sufficiently long, and the plane strain condition is satisfied on cross-sections far from the tunnel face.
- (4) The surrounding rock reaches yield under triaxial compression, and there is no tensile failure zone around the circular tunnel.

2.2. Introduction of GZZ strength criterion

The significant three-dimensional nonlinear strength characteristics are observed during yielding and failure in deep-buried rock masses. This paper introduces the smooth GZZ strength criterion to describe the yielding and failure behavior of deep-buried rock masses. This criterion, modified by Cai et al. (2021) based on the original GZZ (Zhang, 2008; Zhang and Zhu, 2007), overcomes the non-smoothness and non-convexity issues on the π -plane associated with the GZZ criterion. The smooth GZZ, expressed in terms of stress invariants, is formulated as follows:

$$F_{GZZ} = \frac{1}{\sigma_c^{(1/a-1)}} (\sqrt{3J_2})^{1/a} + m_b \left(\frac{\sqrt{3J_2}}{2} - \frac{3J_3}{4J_2} \right) - m_b \frac{I_1}{3} - s\sigma_c = 0 \quad (1)$$

where σ_c represents the uniaxial compressive strength of intact rock; m_b , s , and a are empirical parameters inherited from the Generalized Hoek-Brown (GHB) criterion, reflecting the rock mass quality and structural characteristics; I_1 is the first stress invariant, while J_2 and J_3 are the second and third deviatoric stress invariants, respectively. The expression of I_1 , J_2 , and J_3 are:

$$\begin{cases} I_1 = \sigma_1 + \sigma_2 + \sigma_3 \\ J_2 = \frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_1 - \sigma_3)^2}{6} \\ J_3 = \left(\sigma_1 - \frac{I_1}{3} \right)^2 \left(\sigma_2 - \frac{I_1}{3} \right)^2 \left(\sigma_3 - \frac{I_1}{3} \right)^2 \end{cases} \quad (2)$$

Eq. (1) can also be expressed as stress invariants and stress Lode angle:

$$F_{GZZ} = \frac{1}{\sigma_c^{(1/a-1)}} (\sqrt{3J_2})^{1/a} + \left(\frac{3 + \sin 3\theta_\sigma}{2\sqrt{3}} \right) m_b \sqrt{J_2} - m_b \frac{I_1}{3} - s\sigma_c = 0 \quad (3)$$

where θ_σ represents the stress Lode angle, reflecting the influence of the intermediate principal stress σ_2 on strength.

The GZZ and GHB have the same strength parameters m_b , s , and a . Hoek and Brown (2019) provided the calculation expressions for the strength parameters based on the GSI given by Eq. (4), which comprehensively reflects the blockiness of rock masses, the roughness and filling conditions of joint. At present, these parameters can be rapidly obtained through digital in-situ testing methods (Zhang et al., 2020; Cai et al., 2022a, 2022c).

$$\begin{cases} m_b = m_i \exp[(GSI - 100)/(28 - 14D)] \\ s = \exp[(GSI - 100)/(9 - 3D)] \\ a = 1/2 + [\exp(-GSI/15) - \exp(-20/3)]/6 \end{cases} \quad (4)$$

where m_i is a characteristic parameter of the intact rock, which can be taken as a value according to the suggestions of Hoek and Brown (2019); D is the disturbance parameter that depends upon blasting and stress release.

After years of improvement and refinement, the reliability and applicability of the GZZ strength criterion and its parameter system have been validated in many rock engineering projects (Zhu et al., 2023). However, utilizing the GZZ strength criterion and the complex variable method for the mechanical analysis of surrounding rock poses significant challenges, primarily manifested in:

- (1) The determination of the elastic–plastic boundary in the surrounding rock of a tunnel is a challenging inverse problem in solid mechanics. It remains a formidable task to establish a parameterized representation for the elastic–plastic interface with complex shapes and formulate governing equations amenable to its solution under specific geological conditions.
- (2) In many investigations on tunnel plane-strain mechanics, the influence of tunnel axial stress (i.e., intermediate principal stress) on the stability of surrounding rock has been ignored. Another challenging task lies in considering the intermediate principal stress when employing the complex variable method to address the plastic zone and obtaining a mechanical analysis based on the GZZ three-dimensional strength criterion.

3. Problem description

3.1. Solution of plastic stresses

The post-peak mechanical behavior of hard rocks is characterized by the EBP model, while soft rocks follow the EPP model. Additionally, the post-peak strength remains independent of plastic deformation. Under the condition of small strain, the stress distribution in the plastic zone is statically determined by strength criterion and equilibrium differential equations, independent of the far-field geostress (Hill, 1950; Jaeger et al., 2007; Kachanov, 1971). The Far-field geostress only influences the shape and size of the plastic zone. It is suggested by both theoretical and numerical investigations that, in the absence of shear stress at the inner boundary of a circular tunnel, the plastic zone formed under complex non-hydrostatic stress condition may exhibit a non-circular shape. However, the stress distribution remains axisymmetric within the plastic zone (Lu et al., 2018). Analytical expressions for plastic stress field can be directly derived only when adopting strength criteria characterized by mathematically simplistic forms. For the problem considered in this paper, explicit expressions for each plastic stress components cannot be obtained due to the complexity of the GZZ strength criterion formula (i.e., Eq. (3)). Therefore, the finite difference method is employed to calculate the stress components within plastic zone. The specific solution process is as follows.

For the calculation model considered in this paper, when the shear stress at the inner boundary of the tunnel is free, the consistency condition on the yield surface can be written as:

$$dF = \frac{\partial F}{\partial \sigma_{ij}} d\sigma_{ij} = \frac{\partial F}{\partial \sigma_r} d\sigma_r + \frac{\partial F}{\partial \sigma_\alpha} d\sigma_\alpha + \frac{\partial F}{\partial \sigma_z} d\sigma_z = 0 \quad (5)$$

where σ_r and σ_α are the radial and circumferential stresses in the polar coordinate system, respectively, and σ_z is the out-of-plane stress along the tunnel axis.

Divide the surrounding rock into numerous concentric annular zones, each with a uniform thickness of Δr . Denote the tunnel boundary as $r_{(0)} = R_0$, where the outer radius of the j -th annular zone is given by $r_{(j)} = R_0 + j\Delta r$. Based on the stress boundary conditions at the inner boundary of the tunnel (if there are no support pressure P_i , we have $\sigma_{r(0)} = 0$, $\tau_{r\alpha(0)} = 0$) and Eq. (3), the value of $\sigma_{\theta(0)}$ can be determined.

For the j -th annular zone, employing difference operators as an approximation to replace differential operators, substituting $d\sigma_r \approx \sigma_{r(j)} - \sigma_{r(j-1)}$, $d\sigma_\theta \approx \sigma_{\theta(j)} - \sigma_{\theta(j-1)}$, $d\sigma_z \approx \sigma_{z(j)} - \sigma_{z(j-1)}$ into Eq. (5) yields:

$$\frac{\partial F}{\partial \sigma_{r(j-1)}} (\sigma_{r(j)} - \sigma_{r(j-1)}) + \frac{\partial F}{\partial \sigma_{\alpha(j-1)}} (\sigma_{\alpha(j)} - \sigma_{\alpha(j-1)}) + \frac{\partial F}{\partial \sigma_{z(j-1)}} (\sigma_{z(j)} - \sigma_{z(j-1)}) = 0 \quad (6)$$

where the partial derivatives can be directly computed, and the discretization error can be neglected when the thickness of the annular

zone is sufficiently small.

Within the plastic zone of the surrounding rock, the stress components must satisfy the equilibrium differential equation under an axisymmetric condition:

$$\frac{d\sigma_r}{dr} - \frac{\sigma_\alpha - \sigma_r}{r} = 0 \quad (7)$$

Substituting differences for derivatives, the discrete form of Eq. (7) is obtained as:

$$\frac{\Delta \sigma_r}{r_{(j)} - r_{(j-1)}} - \frac{\sigma_{\alpha(j)} - \sigma_{r(j)}}{r_{(j)}} = 0 \quad (8)$$

For plane strain problems, the axial stress in the tunnel is related to the in-plane stresses by the following equation (Cai et al., 2023):

$$\sigma_z = \nu(\sigma_r + \sigma_\alpha) \text{ In elastic zone} \quad (9a)$$

$$\sigma_z \approx \frac{1}{2}(\sigma_r + \sigma_\alpha) \text{ In plastic zone} \quad (9b)$$

where, ν is the Poisson's ratio of the rock mass.

Thus, it can be observed that when defining normal stress is positive in compression, there is $\sigma_\alpha > \sigma_z > \sigma_r$ within the plastic zone, representing the maximum principal stress σ_1 , intermediate principal stress σ_2 , and minimum principal stress σ_3 , respectively.

Substitute Eq. (8) and Eq. (9b) into Eq. (6) allows the consistency condition to be reformulated as a function with σ_r as the unknown:

$$dF_{GZZ} = f(\sigma_{r(j)}, \sigma_{r(j-1)}) = 0 \quad (10)$$

The numerical solutions for $\sigma_{r(j)}$ can be obtained by solving Eq. (10) once the known value of $\sigma_{r(j-1)}$ is provided. Through iterative procedures, this method enables the solution of stress distribution across the entire plastic zone.

3.2. Solution of elastic stresses

According to the complex variable theory, the stress distribution of surrounding rock in an infinite domain can be given by two analytical functions, denoted as $\varphi_1(z)$ and $\psi_1(z)$. The specific forms of $\varphi_1(z)$ and $\psi_1(z)$ are as follows:

$$\varphi_1(z) = \Gamma z + \sum_{k=1}^{\infty} \alpha_k z^{-k} \quad (11)$$

$$\psi_1(z) = \Gamma' z + \sum_{k=1}^{\infty} \beta_k z^{-k} \quad (12)$$

where $z = x + iy$, $i = \sqrt{-1}$; $\Gamma = (\sigma_x^\infty + \sigma_y^\infty)/4$, $\Gamma' = (\sigma_y^\infty - \sigma_x^\infty)/2$; σ_x^∞ and σ_y^∞ represent the far-field geostresses along the horizontal (x-direction) and vertical (y-direction) axes, respectively.

The values of analytic function coefficients α_k and β_k in Eqs. (11) and (12) are uniquely determined by the boundary conditions. The stress boundary conditions along the inner boundary of the elastic zone in surrounding rock can be represented through the two analytic functions $\varphi_1(z)$ and $\psi_1(z)$ as follows:

$$\varphi(z) + z\overline{\varphi'(z)} + \overline{\psi(z)} = i \oint (X_n + iY_n) ds + C \quad (13)$$

where X_n and Y_n represent the components of the surface forces along the x- and y-axes, respectively, and C is an undetermined complex constant.

The line integral in Eq. (13) is conducted along the boundary of the elastic zone, and the loads on the boundary can be arbitrary. For the problem in this paper, the surface force components on the elastic–plastic boundary can be calculated by the following formula.

$$\begin{cases} X_n = \sigma_x^p l + \tau_{xy}^p m \\ Y_n = \tau_{xy}^p l + \sigma_y^p m \end{cases} \quad (14)$$

where σ_x and σ_y represent the normal stress components along the x and y directions in the Cartesian coordinate system, respectively, and τ_{xy} is the shear stress component. The stress components denoted with the superscript 'p' signify their occurrence within the plastic zone, allowing for a clear distinction from stress components within the elastic zone. Additionally, l and m represent the direction cosines of the unit outward normal $\vec{n}$ at the discussed boundary point. The relationship is expressed as follows:

$$\begin{cases} l = \cos(n, x) = \frac{dy}{ds} \\ m = \cos(n, y) = -\frac{dx}{ds} \end{cases} \quad (15)$$

In the context of localized plastic zone formation around the tunnel, a segmented integration of the right-hand side of Eq. (13) is pertinent. Specifically, the surface force components are represented as $X_n = Y_n = 0$ along the elastic boundary of the tunnel and are defined by Eq. (14) at the elastic-plastic boundary.

Upon determining the specific forms of the analytic functions $\varphi_1(z)$ and $\psi_1(z)$ in accordance with Eq. (13), the stress components within the elastic zone can be represented as follows:

$$\sigma_x^e + \sigma_y^e = 4\text{Re}[\varphi_1'(z)] \quad (16)$$

$$\sigma_y^e - \sigma_x^e + 2i\tau_{xy}^e = 2[\bar{z}\varphi_1''(z) + \psi_1'(z)] \quad (17)$$

where $\text{Re}[\cdot]$ represents taking the real part of the function inside the parentheses, and $\bar{z}$ denotes the conjugate of z .

4. Prediction of plastic zone radius

4.1. Solution of analytic functions

The complexity embedded in stress boundary equation (13) imposes the difficulties on the direct solution of the corresponding analytic function in the physical plane. To overcome this challenge, an initial step is performed involving the application of a conformal mapping within the framework of complex variable method. The mapping is utilized to transform the elastic zone in the physical plane into the outer domain of the unit circle in the image plane. For the two types of plastic zones illustrated in Fig. 1, the elastic zone of the surrounding rock can be regarded as an infinite simply-connected domain, bounded by a non-circular inner boundary (Fig. 2). This holds true regardless of whether the plastic zones completely encircle the circular tunnel. The mapping function can be expressed using a Taylor series as follows:

$$z = \omega(\zeta) = R \left(\zeta + \sum_{k=1}^{\infty} c_k \zeta^{-k} \right) \quad (18)$$

where the complex variable $\zeta = \rho e^{i\theta}$.

The elastic-plastic boundary discussed in this paper is unknown, and thus the mapping function coefficients describing the inner boundary of the elastic zone are unknown variables to be determined. Approaching the problem from a forward analysis perspective, the solution process for the coefficients α_k and β_k in analytic functions $\varphi_1(z)$ and $\psi_1(z)$ will be provided in the following, presuming the mapping function is known. And then this process serves to derive the elastic stress field in the surrounding rock. Substituting $z = \omega(\zeta)$ into the analytic functions $\varphi_1(z)$ and $\psi_1(z)$, the analytic functions can be expressed in terms of ζ as follows:

$$\varphi_1(z) = \varphi_1[\omega(\zeta)] = \varphi(\zeta) = \Gamma\omega(\zeta) + \sum_{k=0}^{\infty} a_k \zeta^{-k} \quad (19)$$

$$\psi_1(z) = \psi_1[\omega(\zeta)] = \psi(\zeta) = \Gamma'\omega(\zeta) + \sum_{k=0}^{\infty} b_k \zeta^{-k} \quad (20)$$

Through the mapping function $z = \omega(\zeta)$, the originally expressed analytic functions in terms of complex variables z can be transformed into analytic functions represented in terms of complex variables ζ , which has superior properties.

Substituting $z = \omega(\zeta)$ into the stress boundary condition (13) and taking the boundary values of ζ as σ , we obtain:

$$\varphi(\sigma) + \frac{\omega(\sigma)}{\omega'(\sigma)} \overline{\varphi'(\sigma)} + \overline{\psi(\sigma)} = i \int_L (X_n + iY_n) ds + C_A \quad (21)$$

where C_A does not affect the stress distribution, so it can be assumed to be zero.

Let the right-hand side of Eq. (21) be $i \int_L (X_n + iY_n) ds = f(\sigma)$. The values of $f(\sigma)$ are all known when the stress within the plastic zone is given. Since $f(\sigma)$ is a function of σ with a period of 2π , it can be expanded into the following Fourier series:

$$f(\sigma) = d_0 + \sum_{k=1}^{\infty} d'_k \sigma^{-k} + \sum_{k=1}^{\infty} d_k \sigma^k \quad (22)$$

where d_0 , d_k , and d'_k are real constants. In the specific computation, the highest positive and negative powers of the Fourier series are both set to n . The solution process of d_0 , d_k , and d'_k can be found in Appendix A.

Similarly, the term $\omega(\sigma)/\omega'(\sigma)$ in the left-hand side of Eq. (21) can be expanded into a Fourier series:

$$\frac{\omega(\sigma)}{\omega'(\sigma)} = \sum_{k=1}^m e'_k \sigma^{-k} + \sum_{k=1}^{\infty} e_k \sigma^k + e_0 \quad (23)$$

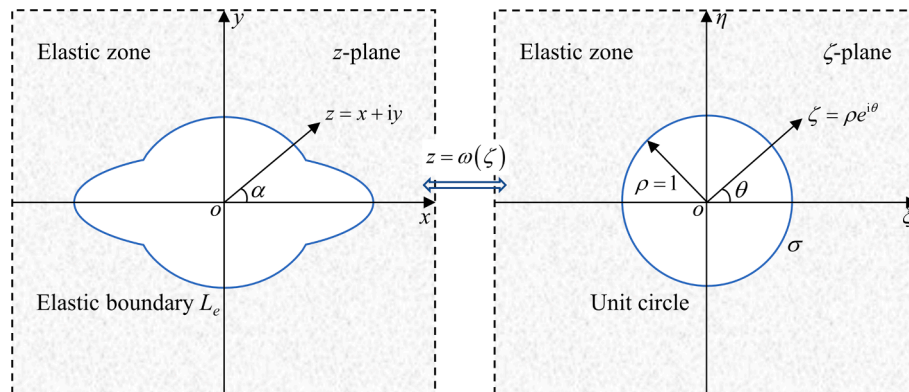


Fig. 2. The conformal mapping from the infinite simply-connected domain on the physical plane to the outer domain of the unit circle on the image plane.

where e_0 , e_k , and e'_k are determined by the mapping function coefficients, that is:

$$e'_m = c_m$$

$$e'_{m-1} = c_{m-1}$$

$$e'_{m-j+1} = \sum_{k=1}^{j-2} (j-1-k)c_{j-1-k}e'_{m-k+1} + c_{m-j+1}j = 3, 4, \dots, m$$

$$e_0 = \sum_{k=1}^{m-1} kc_k e'_{k+1}$$

$$e_1 = \sum_{k=1}^m kc_k e'_k + 1$$

$$e_j = \sum_{k=j}^m kc_k e'_{k-j+1} + \sum_{k=1}^{j-1} kc_k e_{j-k-1} (j = 2, \dots, m)$$

Substituting Eqs. (18) and (22) into the right-hand side of Eq. (24), we obtain:

$$f_1(\sigma) = d_0 + \sum_{k=1}^n d'_k \sigma^{-k} + \sum_{k=1}^n d_k \sigma^k - 2\Gamma R \left(\sigma + \sum_{k=1}^m c_k \sigma^{-k} \right) - \Gamma' R \left(\sigma^{-1} + \sum_{k=1}^m c_k \sigma^k \right) \quad (25)$$

Substituting Eq. (23) into the left-hand side of Eq. (24), we obtain:

$$a_0 + b_0 + \sum_{k=1}^{\infty} a_k \sigma^{-k} - \left(\sum_{k=1}^m e'_k \sigma^{-k} + \sum_{k=1}^{\infty} e_k \sigma^k + e_0 \right) \sum_{k=1}^{\infty} ka_k \sigma^{k+1} + \sum_{k=1}^{\infty} b_k \sigma^k = f_1(\sigma) \quad (26)$$

where only the value of $a_0 + b_0$ affects the surrounding rock stress, and thus we take $b_0 = 0$.

Then, by comparing the terms with the same powers of σ in Eqs. (25) and (26), we obtain:

$$\left\{ \begin{array}{ll} \sigma^{-1} : & a_1 - \sum_{k=3}^m (k-2)e'_k a_{k-1-1} = d'_1 - 2\Gamma R c_1 - \Gamma' R \\ \sigma^{-j} (2 \leq j \leq m-2) : & a_j - \sum_{k=j+2}^m (k-j-1)e'_k a_{k-j-1} = d'_j - 2\Gamma R c_j \\ \sigma^{-(m-1)} : & a_{m-1} = d'_{m-1} - 2\Gamma R c_{m-1} \\ \sigma^{-m} : & a_m = d'_m - 2\Gamma R c_m \\ \sigma^{-j} (m+1 \leq j \leq n) : & a_j = d'_j \\ \sigma^0 : & a_0 - \sum_{k=2}^m (k-1)e'_k a_{k-1} = d_0 \\ \sigma^1 : & -\sum_{k=1}^m k e'_k a_k + b_1 = d_1 - 2\Gamma R - \Gamma' R c_1 \\ \sigma^j (2 \leq j \leq m) : & -\sum_{k=0}^{j-2} (j-k-1)e_k a_{j-k-1} - \sum_{k=1}^m (j+k-1)e'_k a_{j+k-1} + b_j = d_j - \Gamma' R c_j \\ \sigma^j (m+1 \leq j \leq n-m+1) : & -\sum_{k=0}^{j-2} (j-k-1)e_k a_{j-k-1} - \sum_{k=1}^m (j+k-1)e'_k a_{j+k-1} + b_j = d_j \\ \sigma^j (n-m+2 \leq j \leq n) : & -\sum_{k=0}^{j-2} (j-k-1)e_k a_{j-k-1} - \sum_{k=1}^{n+1-j} (j+k-1)e'_k a_{j+k-1} + b_j = d_j \\ \sigma^{n+1} : & -\sum_{k=0}^{n-1} (n-k)e_k a_{n-k} + b_j = 0 \\ \sigma^j (n+2 \leq j \leq 2n+1) : & -\sum_{k=j-n-1}^n (j-k-1)e_k a_{j-k-1} + b_j = 0 \end{array} \right. \quad (27)$$

$$e_j = \sum_{k=1}^m kc_k e_{j-k-1} (j = m+1, \dots, \infty)$$

According to Eqs. (19)-(23), we obtain:

$$a_0 + b_0 + \sum_{k=1}^{\infty} a_k \sigma^{-k} - \frac{\omega(\sigma)}{\omega'(\sigma)} \sum_{k=1}^{\infty} ka_k \sigma^{k+1} + \sum_{k=1}^{\infty} b_k \sigma^k = f_1(\sigma) \quad (24)$$

$$f_1(\sigma) = f(\sigma) - 2\Gamma \omega(\sigma) - \Gamma' \overline{\omega(\sigma)}$$

From Eq. (27), we can obtain a system of $3n+2$ linear equations with respect to the $3n+2$ unknown coefficients $a_k (k = 0, 1, 2, \dots, n)$ and $b_k (k = 1, 2, \dots, 2n+1)$ of the two analytic functions to be solved. Therefore, directly solving the system of linear equations yields the specific expressions for the two analytic functions $\varphi(\zeta)$ and $\psi(\zeta)$. Subsequently, the stress combinations in the elastic zone of the surrounding rock are obtained by substituting $z = \omega(\zeta)$, $\varphi(\zeta)$, and $\varphi(\zeta)$ into Eqs. (16) and (17).

$$\sigma_x^e + \sigma_y^e = 4\text{Re}[\Phi(\zeta)] \quad (28)$$

$$\sigma_y^e - \sigma_x^e + 2i\tau_{xy}^e = 2 \left[\overline{\omega(\zeta)} \frac{\Phi'(\zeta)}{\omega'(\zeta)} + \Psi(\zeta) \right] \quad (29)$$

$$\text{where } \Phi(\zeta) = \frac{\phi'(\zeta)}{\omega'(\zeta)} \text{ and } \Psi(\zeta) = \frac{\psi'(\zeta)}{\omega'(\zeta)}.$$

Then, the stress components in the elastic region of the surrounding rock can be obtained from Eqs. (28) and (29), that is:

$$\sigma_x^e = 2\text{Re}[\Phi(\zeta)] - \text{Re} \left[\overline{\omega(\zeta)} \frac{\Phi'(\zeta)}{\omega'(\zeta)} + \Psi(\zeta) \right] \quad (30)$$

$$\sigma_y^e = 2\text{Re}[\Phi(\zeta)] + \text{Re} \left[\overline{\omega(\zeta)} \frac{\Phi'(\zeta)}{\omega'(\zeta)} + \Psi(\zeta) \right] \quad (31)$$

$$\tau_{xy}^e = \text{Im} \left[\overline{\omega(\zeta)} \frac{\Phi'(\zeta)}{\omega'(\zeta)} + \Psi(\zeta) \right] \quad (32)$$

where $\text{Im}[\cdot]$ represents taking the imaginary part of the function inside the parentheses $[\cdot]$.

4.2. Solution of plastic zone boundary

The strength criterion, employing the stress state of the rock mass as a parameter, explores its correlation with failure. Consequently, substituting Eq. (9a) and Eqs. (30)–(32) into the strength criterion (3) yields the function F_{GZZ} , which characterizes the relationship between the stress state at any given point in the surrounding rock and the critical stress state. The tendency of F_{GZZ} approaching zero indicates a stress state nearing the critical limit, suggesting an elevated susceptibility to failure. And it indicates that the material has attained either yielding or failure when F_{GZZ} equals zero. It can be expressed as:

$$\begin{cases} F_{GZZ}(I_1, J_2, \theta_\sigma) = 0 & \text{In plastic state} \\ F_{GZZ}(I_1, J_2, \theta_\sigma) < 0 & \text{In elastic state} \end{cases} \quad (33)$$

However, Eq. (33) fails to capture the influence of the stress level on the yield or failure of the rock mass. Therefore, it is possible to introduce the dimensionless failure approach index FAI_{GZZ} to quantitatively evaluate the failure degree of a rock element. As illustrated in Fig. 3, the stress state of a rock element is represented by point P , and FAI_{GZZ} reflects the relative position of the stress state concerning its strength envelope. On the π -plane, it can be located either on or inside the strength envelope (Cai et al., 2022b). When point P is located inside the smooth GZZ envelope, indicating that the rock has not failed. FAI_{GZZ} can be defined as:

$$\begin{aligned} FAI_{GZZ} &= \frac{\overline{OP}}{\overline{OQ}} = \frac{\tau_{oct}(\sigma_1, \sigma_2, \sigma_3)}{\tau_{oct}(I_1, \theta_\sigma)} = \frac{\sqrt{2J_2/3}}{\tau_{oct}(I_1, \theta_\sigma)} \\ &= \frac{\sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_1 - \sigma_3)^2}}{3\tau_{oct}(I_1, \theta_\sigma)} \end{aligned} \quad (34)$$

where $\overline{OQ} = \tau_{oct}(I_1, \theta_\sigma)$ represents the radius of the smooth GZZ strength envelope on the π -plane, which can be obtained by solving the GZZ strength criterion (3). When $FAI_{GZZ} = 1$, points P and Q coincide, indicating the rock is in a limit failure state. Under the condition of hydrostatic pressures ($\sigma_1 = \sigma_2 = \sigma_3$), i.e., when point P is located at the origin O , it means that the rock does not undergo yielding or failure. In summary, we have:

In summary, we have:

$$\begin{cases} FAI_{GZZ}(I_1, J_2, \theta_\sigma) = 1 & \text{In plastic state} \\ FAI_{GZZ}(I_1, J_2, \theta_\sigma) < 1 & \text{In elastic state} \end{cases} \quad (35)$$

Additionally, regardless of whether the plastic zone only locally develops around the tunnel or completely encircles the tunnel, the elastic zone of the surrounding rock always extends beyond the tunnel section. Therefore, the geometric conditions that the elastic zone must satisfy are:

$$r(z) = \sqrt{\omega(\zeta)\overline{\omega(\zeta)}} > R_0 \quad (36)$$

In the following, we turn our attention to establishing the boundary conditions for determining the distribution of the plastic zone. Let z_j represent any arbitrary point on the inner boundary L_e of the elastic zone, corresponding to a point σ_j on the image plane. Under the action of in-situ stress, the surrounding rock is in a state of limit equilibrium. For the EBP model, the surrounding rock exists in a residual strength state in the inner side of the elastic–plastic boundary L , with every point in a plastic state, as illustrated by point D in Fig. 4. On the outer side of L , the surrounding rock is at its peak strength, and the stress distribution is in a pre-peak state, represented by points A , B , and C in Fig. 4. In summary, any point σ_j on L_e should satisfy Eqs. (35) (or (33)) and (36). Additionally, the surrounding rock fully utilizes its peak strength along the outer side of elastic–plastic boundary L . That is, the stress state at any point on the outer side L of should strive to closely approach the peak stress state, specifically as close as possible to point C in Fig. 4. This means that F_{GZZ} tends towards 0 and FAI_{GZZ} tends towards 1.

Hence, it is evident that the stresses along the elastic–plastic boundary may experience discontinuities. More precisely, the stress values might exhibit a discontinuous distribution along the tangential direction of the elastic–plastic boundary, which will be demonstrated in the following examples. Specifically, the peak strength of the

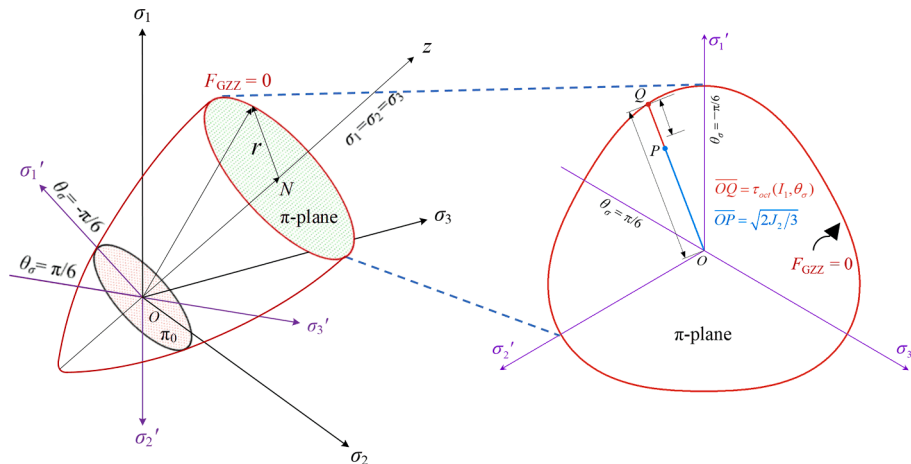


Fig. 3. The stress state of the rock and the smooth GZZ strength yield surface on the π -plane.

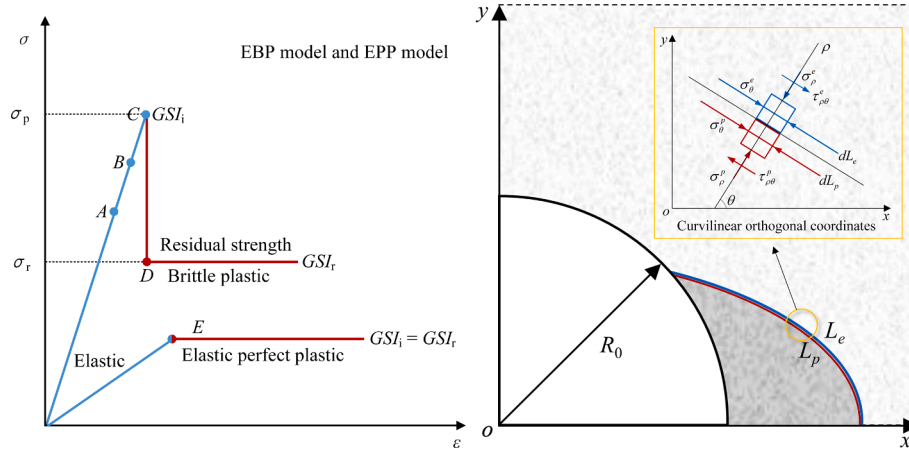


Fig. 4. Schematic diagram of elastic and plastic stress components on both sides of the elastic-plastic boundary under limit equilibrium conditions.

surrounding rock can be considered equal to the residual strength for the EPP model. And the surrounding rocks have all reached the plastic state on the outer side of the elastic-plastic boundary, corresponding to point E in Fig. 4.

This paper will employ optimization techniques to solve problems of elastic-brittle-plastic nature. It involves describing the mapping function coefficients of the inner boundary of the elastic zone in the surrounding rock as design variables X . The objective is to find the values X^* that minimize the objective function $f(X)$, subject to the inequality constraints $g_j(X)$ and $h_j(X)$. The mathematical model is given by:

$$\begin{cases} \min f(X) \sigma_j \in L_p \\ s.t. g_j(X) \leq 0 \sigma_j \in L_e \\ h_j(X) \geq 0 \sigma_j \in L_e \end{cases} \quad (37)$$

where $X \in (R, c_1, c_2, \dots, c_m)$, $f(X) = \sum f_j(X)$ is the objective function of the optimization problem, and $f_j(X) = [1 - FAI_{GZZ}(I_1, J_2, \theta_\sigma)]_{z_j}^2$. The magnitude of $f(X)$ reflects the strength exerted by the surrounding rock on the outer side of elastic-plastic boundary. A larger value indicates higher strength redundancy in the surrounding rock, making it more stable. Conversely, a smaller value suggests that the surrounding rock is more prone to failure. And $g_j(X) = 1 - FAI_{GZZ}(I_1, J_2, \theta_\sigma)|_{z_j}$, $h_j(X) = r(z_j) - R_0$.

Therefore, based on Eq. (37), the penalty function method is employed to integrate the inequality constraint functions $g_j(X)$ and $h_j(X)$ into the objective function $f(X)$. This leads to the construction of a new unconstrained augmented objective function $F(X)$, given by:

$$F(X) = f(X) + r_1 \sum_{\sigma_j \in L} (\max[0, g_j(X)])^2 + r_2 \sum_{\sigma_j \in L} (\min[0, h_j(X)])^2 \quad (38)$$

where r_1 and r_2 represent penalty factors associated with the inequality constraint functions $g_j(X)$ and $h_j(X)$. In practical calculations, these factors are conventionally set to significant magnitudes.

In this manner, the constrained minimization problem for the original objective function $f(X)$ is reformulated as an unconstrained minimization problem for the penalty function $F(X)$ using Eq. (38). The optimization problem is then solved by the differential evolution algorithm in this study.

Specifically, the strength of the surrounding rock on the elastic-plastic boundary remains constant and all stress components of the surrounding rock exhibit continuity for the EPP model. Consequently, Eq. (38) attains a rigorous minimum value of zero. Many scholars (Ghorbani and Hasanadehshooili, 2019; Lu et al., 2018; Xu and Xia, 2021) have studied this elastic-perfect-plastic problem, and our study comprehensively encompasses this specialized case. In essence, Eq. (38)

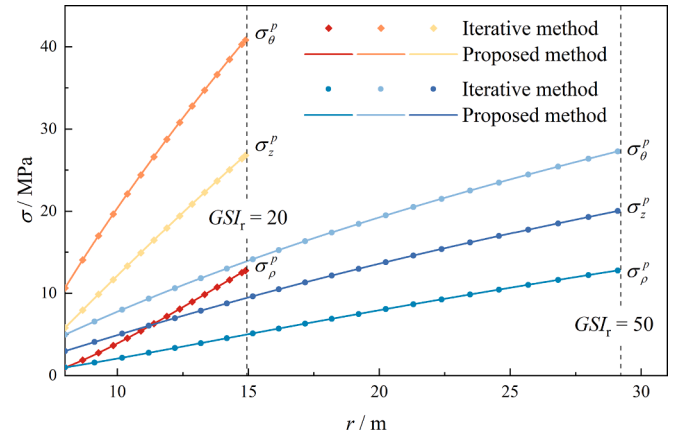


Fig. 5. The radii and stresses of the plastic zone obtained by the two methods for different GSI_r .

is applicable to both cases where the plastic zone is generated locally around the tunnel and completely surrounds a circular tunnel.

5. Verification and validation

5.1. Verification of plastic stresses

To rigorously verify the theoretical derivations in this paper, the results obtained by our method will be compared with the existing research findings. Su et al. (2022) proposed a semi-analytical solution for the elastic-brittle-plastic behavior of circular tunnels based on the GZZ strength criterion. However, the solution is limited to axisymmetric problems. They employed an improved iterative method to determine the stress distribution in the plastic zone. In contrast, the method proposed in this paper requires no iterations, resulting in higher computational efficiency. The solution presented in this paper can degenerate into an axisymmetric solution, if the far-field horizontal stress and vertical stress are set to be equal.

The calculation parameters are set as: $R_0 = 8\text{m}$, $\sigma_x^\infty = \sigma_y^\infty = 30\text{MPa}$, $\sigma_c = 40\text{MPa}$, $m_i = 6$, $\nu = 0.35$, $D = 0$, $GSI_i = 60$, $P_i = 1\text{MPa}$. Fig. 5 illustrates the comparison of the plastic zone radius and plastic stress in the surrounding rock when GSI_r is set to 20 and 50, respectively. The stress distribution and plastic zone radius characteristics calculated in this paper are in substantial agreement with the results obtained by Su et al. (2022), indicating the correctness of the theoretical derivations in this study.

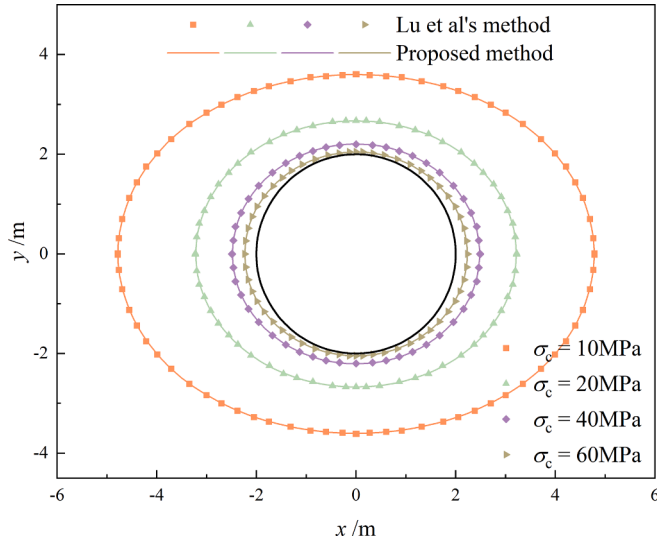


Fig. 6. The comparison of the elastic–plastic boundaries obtained by two methods in a non-axisymmetric stress field.

5.2. Comparison with non-axisymmetric problem

When employing the two-dimensional HB strength criterion, it is possible to directly derive analytical expressions for the stress in the plastic zone (Osgoui and Oreste, 2010; Sharan, 2008; Tu et al., 2018; Zhang et al., 2016). Lu et al. (2018) provided a plane elastic–plastic analytical solution considering a non-axisymmetric stress field while neglecting the axial stress. To show the applicability of the proposed method, a comparative verification is conducted by replacing only the GZZ criterion (1) with the two-dimensional HB criterion.

The calculation parameters are set as: $R_0 = 2\text{m}$, $\sigma_y^\infty = 30\text{MPa}$, $m_i = 10$. Fig. 6 depicts the elastic–plastic boundaries when the uniaxial compressive strengths of intact rock are 10 MPa, 20 MPa, 40 MPa, and 60 MPa, respectively. The comparative results demonstrate a high level of agreement between the elastic–plastic boundaries obtained by the two methods, thereby thoroughly validating the accuracy of the theoretical derivations in this paper.

5.3. Validation of 3D strength characteristics consideration

Ignoring the 3D strength characteristics of the surrounding rock will result in an overestimation of the plastic zone, thereby imparting a conservative support design (Xu and Yu, 2006; Zhou and Li, 2011; Singh et al., 2019; Ghorbani and Hasanzadehshooili, 2019; Xu and Xia, 2021). The analytical solution provided by Lu et al. (2018) is also applicable to the GHB criterion; a researcher can substitute the expressions for plastic stress with:

$$\sigma_r = \frac{[s^{1-a} + (1-a)m_b \ln(r/R_0)]^{1/(1-a)} - s}{m_b} \sigma_c \quad (39)$$

$$\sigma_\alpha = \sigma_r + \sigma_c \left(m_b \frac{\sigma_r}{\sigma_c} + s \right)^a \quad (40)$$

The GHB criterion neglects the 3D strength characteristic of the rock mass, a consideration accurately accounted for by the GZZ criterion. Therefore, we compared the plastic zones obtained through these two criteria and analyzed the differences in plastic zone characteristics within a non-hydrostatic stress field.

The calculation parameters are set as follows: $\sigma_y^\infty = 40\text{MPa}$, $GSI = 60$, $\nu = 0.3$, the lateral pressure coefficient λ takes values of 0.5, 1.0, and 1.50, respectively. The remaining parameters are identical to those in Section 5.2. The comparative results obtained are shown in Fig. 7. Under

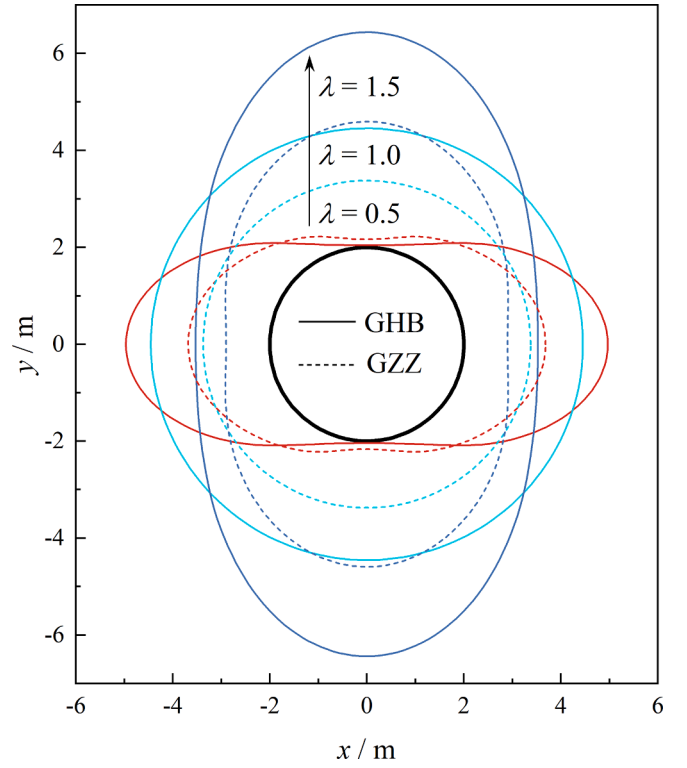


Fig. 7. The influence of 3D strength characteristic of the rock mass under various lateral pressure coefficients on the plastic zone.

the condition of non-hydrostatic stress, it can be observed that the plastic zone significantly decreases in the direction of the minimum in-situ stress, while the opposite is true in the direction of the maximum in-situ stress. Specifically, when $\lambda = 0.5$, the plastic zone actually increases; however, overall, the plastic zone area effectively decreases. Accurately considering the 3D strength characteristics of the rock mass can significantly reduce the maximum thickness $r_{\max}$ of the plastic zone, promoting uniformity in the thickness of the plastic zone. Therefore, it is imperative to simultaneously account for the complex true three-dimensional stress environment in which the tunnel exists and the 3D strength characteristics inherent to the rock mass, when conducting stability assessments of surrounding rock.

6. Discussion and analysis

The morphological characteristics and evolution laws of the plastic zone in surrounding rock around underground tunnels are influenced by various factors, including the in-situ stress environment, rock properties, and the development of joints. To systematically investigate the impact of various factors on the plastic zone, this study conducted an extensive analysis of numerous cases. Following the work made in references (Martin et al., 1999; Perras and Diederichs, 2016), the depth of the plastic zone is normalized with respect to the radius of the circular tunnel and the geostress is normalized with respect to the uniaxial compressive strength σ_c of intact rock.

6.1. Plastic zone characteristics in soft rock

6.1.1. Influence of λ

Consider a circular tunnel excavated in deep soft rock mass, with a radius of R_0 . The mechanical behavior during the elastic–plastic stage can be described using the EPP model. This paper adopts the surrounding rock parameters as follows: $\sigma_c = 40\text{MPa}$, $m_i = 8$, $\nu = 0.3$, $D = 0$, $GSI = 40$. In the context of a fixed vertical in-situ stress of $\sigma_y^\infty =$

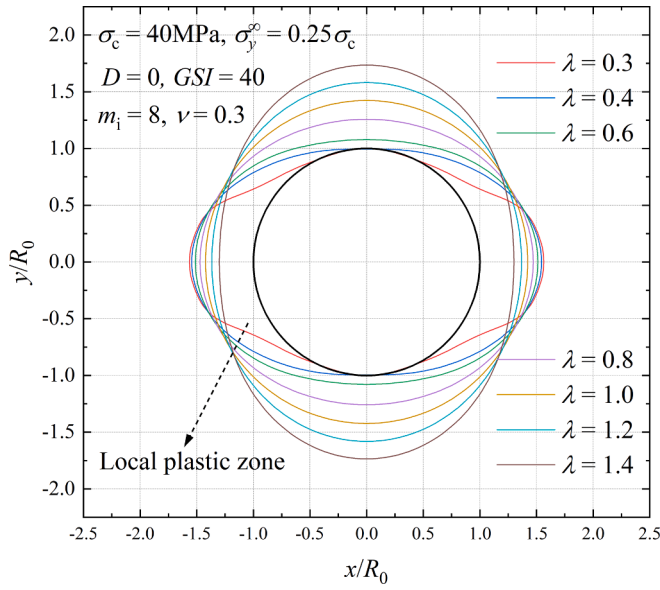


Fig. 8. Plastic zones around a deep buried circular tunnel in soft rock under varied lateral pressure coefficients.

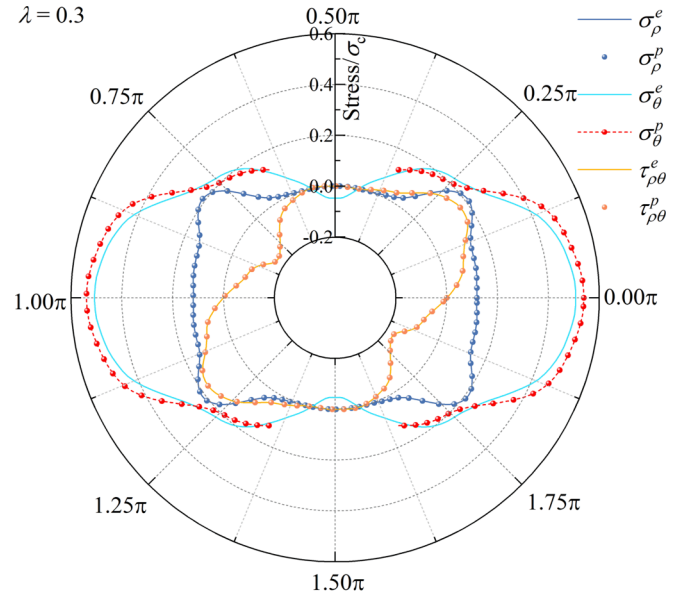
$0.25\sigma_c$, the evolution of the morphological characteristics of the plastic zone under various horizontal stress conditions was investigated. The influence of the lateral pressure coefficient $\lambda = \sigma_x^\infty/\sigma_y^\infty$ on the development of the plastic zone was investigated, and the results are shown in Fig. 8.

Fig. 8 illustrates that the lateral pressure coefficient significantly influences the morphological characteristics of the plastic zone, consistently reaching maximum depth in the direction of the smaller horizontal geostress. The lower uniaxial compressive strength of the soft rock results in a substantial plastic zone even under lower stress levels (in this case, with a maximum horizontal stress of 14 MPa and a fixed vertical stress of 10 MPa). The plastic zone is confined to a localized region and exhibits a U-shaped morphology in the surrounding rock when the relative difference between the in-situ stresses along the two direction is significant, as indicated by the red solid line in Fig. 8. In this case, the in-situ stress difference accounts for 70 % of σ_y^∞ and 17.5 % of σ_c .

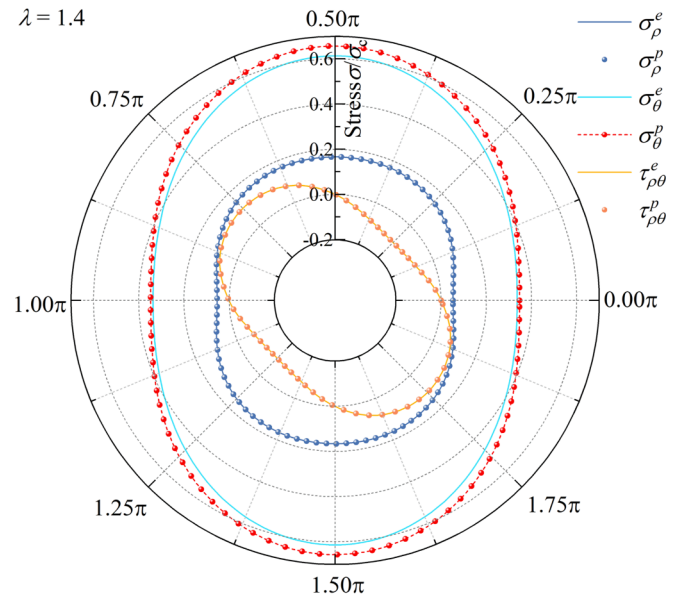
From the analysis in Section 3.1, it becomes evident that, even when incorporating the EPP model, Eq. (9) may result in an inability to maintain full continuity of stress components along the elastic–plastic interface. Moreover, this discontinuity becomes more pronounced with a reduction in the Poisson’s ratio of rock mass. To investigate the errors introduced by Eq. (9), Fig. 9 shows the in-plane stress components on the elastic–plastic interface for different values of λ , specifically, 0.3 and 1.4. The continuity of each stress component is observed for scrutiny. It is important to note that these stress components are presented in an orthogonal curvilinear coordinate system determined by the mapping function, as defined in Eq. (18).

From Fig. 9, it can be observed that under the non-hydrostatic stress condition, the tangential stress component σ_θ on the elastic–plastic interface is discontinuous. The other two stress components σ_ρ and $\tau_{\rho\theta}$ remain continuous. Notably, the error between the elastic tangential stress σ_θ^e and the plastic tangential stress σ_θ^p is small. This indicates that the consideration of Eq. (9) introduces errors into the calculation results within an acceptable range.

In particular, Fig. 9(a) indicates that when there is a significant difference between the in-situ stresses along horizontal and vertical direction, not only does the plastic zone fail to completely enclose the circular tunnel, but also a complex stress state involving tensile tangential stresses ($\sigma_\theta < 0$) emerges at the top and bottom of the tunnel.



(a)



(b)

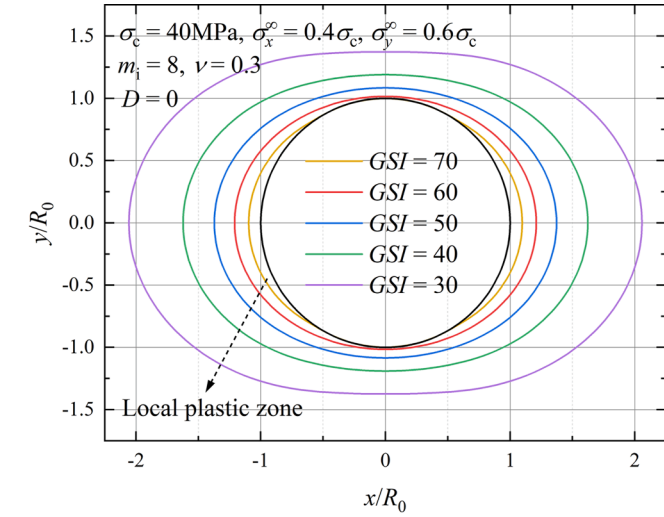
Fig. 9. The elastic and plastic stress components on the inner boundary of the elastic zone in the surrounding rock: (a) $\lambda = 0.3$; (b) $\lambda = 1.4$.

This condition is highly prone to inducing tensile failure, detrimentally affecting the surrounding rock stability.

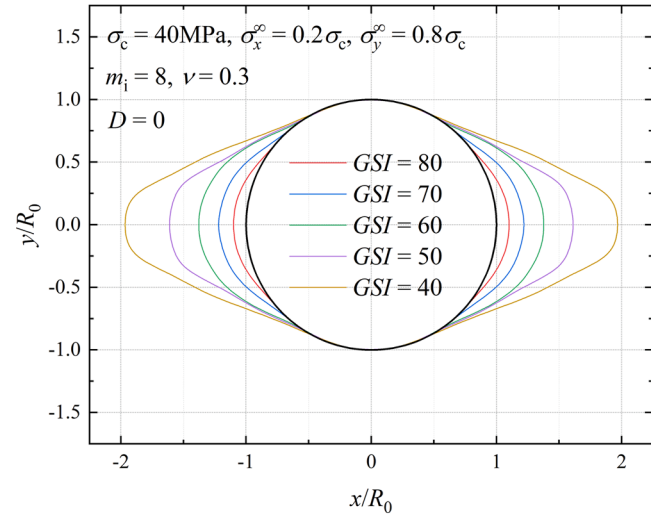
6.1.2. Influence of GSI

The GSI reflects the integrity of surrounding rock. Numerous studies (Xu and Xia, 2021; Su et al., 2022, 2023) indicate that the GSI value significantly influences the assessment of plastic zone size. However, these studies have been limited to axisymmetric stress conditions. This section explores the influence of different GSI values on the plastic zone under two lateral pressure coefficient cases: $\lambda = 2/3$ and $\lambda = 1/4$, representing distinct in-situ stress environments. The other parameters remain consistent with those described in Section 6.1.1. The results are plotted in Fig. 10.

Fig. 10(a) reveals that when the difference between the horizontal



(a)



(b)

Fig. 10. The distribution of the plastic zone for different GSI values: (a) $\lambda = 2/3$; (b) $\lambda = 1/4$.

and vertical in-situ stresses is relatively small (with λ close to 1), the plastic zone typically encloses the entire circular tunnel. Furthermore, when the GSI value is relatively high, indicating a more intact rock mass, the surrounding rock near the tunnel wall yields along the horizontal direction, resulting in small, crescent-shaped plastic zones. Fig. 10(b) indicates that the plastic zone fails to fully enclose the circular tunnel when there is a large difference in the horizontal and vertical in-situ stresses (with small λ). As the GSI decreases, the plastic zone undergoes a morphological transformation from a crescent shape to a U-shape. This variation stands out distinctly from the scenario characterized by a slight difference in the in-situ stresses. Moreover, the tangential stresses σ_θ at the top and bottom of the rock masses around the tunnel exhibit negative values, indicating tensile states in the tangential direction at these locations. When GSI = 80, the absolute value of the maximum tangential tensile stress is $0.044\sigma_c$. As GSI decreases, the maximum tangential stress continues to decrease. When GSI = 30, the absolute value of the maximum tangential tensile stress is $0.096\sigma_c$. This implies that when excavating a circular tunnel in geological environments with a significant difference in in-situ stress along the two directions, the surrounding rock with well-developed joints is not only

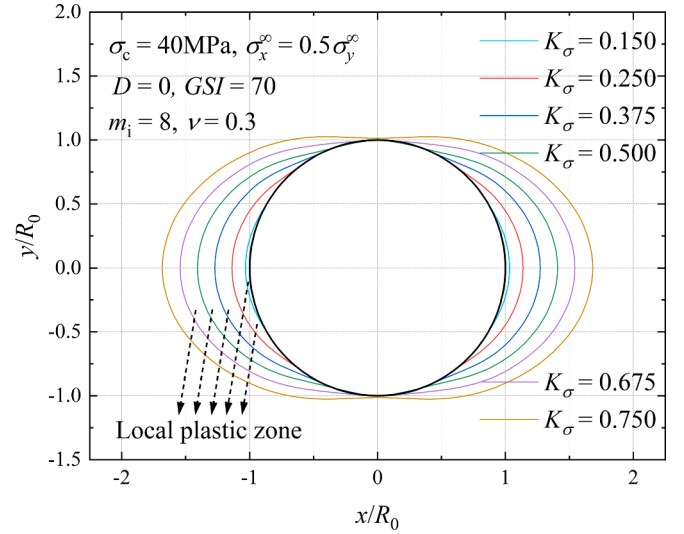


Fig. 11. The distribution of the plastic zone under different values of K_σ .

prone to developing a large plastic zone but also exhibits a tendency for tensile failure along the direction of higher in-situ stress.

6.1.3. Influence of in-situ stress level

For tunnels excavated in deep soft rock masses, the strength of intact rock is relatively low compared to the in-situ stress level. Therefore, some scholars (Martin et al., 1999) compiled experience from tunnels buried at different burial depths and suggested the assessment of tunnel stability by using the ratio of far-field maximum stress (σ_x^∞ or σ_y^∞) to the uniaxial compressive strength σ_c of intact rock.

In this study, we define the geostress-to-strength ratio K_σ as the ratio of the average value of horizontal and vertical in-situ stresses to uniaxial compressive strength of the rock. And then, the utilization of K_σ to assess the in-situ stress level on the stability of tunnel surrounding rock is suggested. The expression for K_σ is given by:

$$K_\sigma = \frac{\sigma_x^\infty + \sigma_y^\infty}{2\sigma_c} \quad (41)$$

Subsequently, an exploration was conducted into the variations in the plastic zone under different K_σ values. The lateral pressure coefficient is fixed at 0.5 and the degree of joint development in the surrounding rock keeps constant ($GSI = 70$). The other parameters are consistent with those in Section 6.1.1. The results are presented in Fig. 11.

It can be seen from Fig. 11 that K_σ predominantly influences the extent of the plastic zone within a fixed lateral pressure coefficient. In the case of the relatively intact surrounding rock under consideration, a small K_σ leads to the plastic zone developing only in a localized area around the circular tunnel. It is only when K_σ is significantly large ($K_\sigma \geq 0.750$) that the plastic zone can completely enclose the circular tunnel. As K_σ increases, the plastic zone maintains a relatively constant rate of development in both the horizontal and vertical directions. This suggests that, despite the expansion of the plastic zone, its boundary shape remains almost unchanged, and the thickness of the plastic zone increases uniformly.

6.1.4. The evolution law of r_{max}

The stress-induced yielding or failure behavior of the surrounding rock typically initiates at the location of maximum stress concentrations near the tunnel wall. Then, it gradually evolves into a distinct plastic zone within its vicinity as the loads increase. For an unsupported circular tunnel, the stress concentration factor K_{SCF} at the free face can be defined as (Martin et al., 1999):

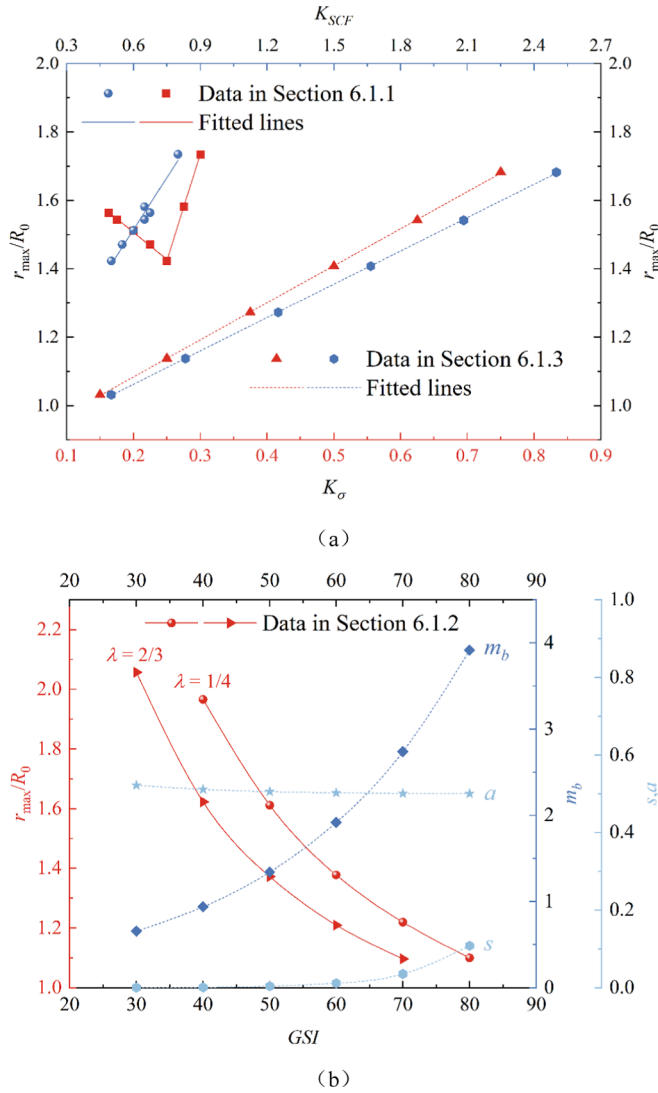


Fig. 12. The evolution law of the maximum thickness $r_{\max}$ of plastic zone for varying: (a) K_{σ} and K_{SCF} ; (b) GSI .

$$K_{SCF} = \begin{cases} \frac{3\sigma_x^{\infty} - \sigma_y^{\infty}}{2\sigma_c}, & \sigma_x^{\infty} \geq \sigma_y^{\infty} \\ \frac{3\sigma_y^{\infty} - \sigma_x^{\infty}}{2\sigma_c}, & \sigma_x^{\infty} < \sigma_y^{\infty} \end{cases} \quad (42)$$

where K_{SCF} represents the normalized value of the maximum tangential stress at the free face of a circular tunnel, with respect to the uniaxial compressive strength of intact rock.

The maximum thickness $r_{\max}$ of the plastic zone in the surrounding rock has a significant impact on tunnel support design. Firstly, the relationship among K_{σ} , K_{SCF} , and $r_{\max}$ is explored using the data from Sections 6.1.1 and 6.1.3. Linear fitting using the least squares method was then performed on the data. The data covers cases where the plastic zone cannot completely enclose a circular tunnel, as well as cases involving non-hydrostatic in-situ stresses and varying lateral pressure coefficients. The results are shown in Fig. 12(a). Similarly, the data in Section 6.1.3 were used to explore the relationship between GSI and $r_{\max}$ when keeping K_{σ} and K_{SCF} constant. The results are shown in Fig. 12(b).

The blue spheres and regular hexagons in Fig. 12(a), drawn from the data in Section 6.1.1 and Section 6.1.3, respectively, represent different lateral pressure coefficients and in-situ stress levels. They reflect the evolution law of $r_{\max}$ around a circular tunnel under various complex stress conditions. It can be observed from Fig. 12(a) that there is a

significant relationship between $r_{\max}$ and K_{SCF} : $r_{\max}$ linearly increases with the increment of K_{σ} . This linear law remains consistent when the strength of the surrounding rock and the degree of joint development are held constant. Importantly, this law is independent of the relative magnitude of far-field stresses and the stress level experienced by the surrounding rock.

The red square in Fig. 12(a) is plotted based on the data from Section 6.1.1, where the minimum value of $r_{\max}$ aligns with λ equaling 1. Through segmented fitting of $r_{\max}$, it is observed that $r_{\max}$ linearly decreases with increasing K_{σ} when λ is less than 1; and $r_{\max}$ linearly increases with K_{σ} when λ is greater than 1, exhibiting a more rapid rate of increase. This implies that the hydrostatic stress field contributes to minimizing the maximum thickness of the plastic zone around a circular tunnel. Conversely, when the value of λ deviates from 1 (that is, there is a greater difference between horizontal and vertical in-situ stress), the plastic zone is more likely to develop deeper into the surrounding rock, posing a significant disadvantage to the stability of rock mass. The red triangle in Fig. 12(a) is plotted based on the data from Section 6.1.3. Through linear fitting, it was observed that $r_{\max}$ increases monotonically with K_{σ} when λ is given. This indicates that the magnitude of K_{σ} can intuitively reflect the stability of the unsupported circular tunnel surrounding rock under complex stress conditions.

In summary, when considering the excavation of a circular tunnel in deep rock masses, the maximum thickness $r_{\max}$ of the plastic zone can be linearly estimated using the geostress-to-strength ratio K_{σ} and the stress concentration factor K_{SCF} . The law about K_{σ} is related to the magnitude of lateral pressure coefficient λ , but the monotonic linear law still holds when λ is fixed.

Fig. 12(b) demonstrates that the magnitude of the GSI significantly influences $r_{\max}$. Under different lateral pressure coefficients, $r_{\max}$ exhibits a nonlinear decrease with increasing GSI . This effect is particularly pronounced when GSI is low, as increasing GSI can markedly reduce $r_{\max}$.

6.2. Plastic zone characteristics in hard rock

6.2.1. Influence of strength drop

When considering a circular tunnel excavated in deep hard rock, the mechanical behavior during the elastic-plastic stage can be described using EBP model. The parameters are: $\sigma_c = 100\text{MPa}$, $m_i = 15$, $\nu = 0.3$, $D = 0$, $\sigma_x^{\infty} = 0.25\sigma_c$, $\sigma_y^{\infty} = 0.50\sigma_c$. This study explored the impact of varying strength drops in rock mass on the development of the plastic

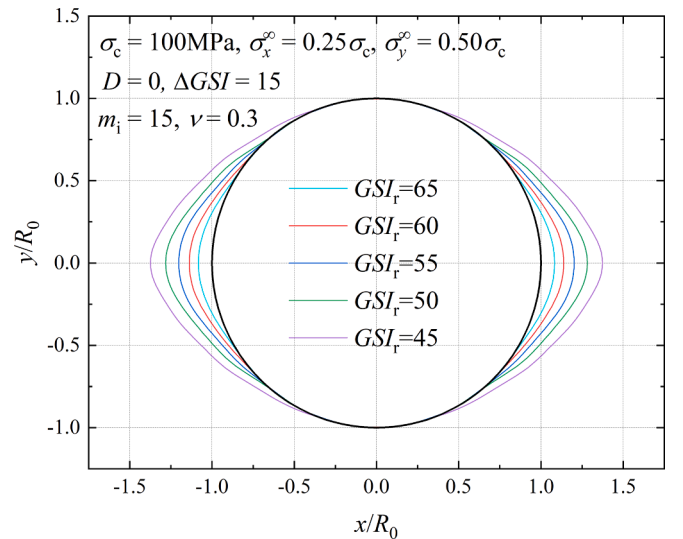


Fig. 13. The plastic zone within the surrounding rock influenced by different degrees of joints.

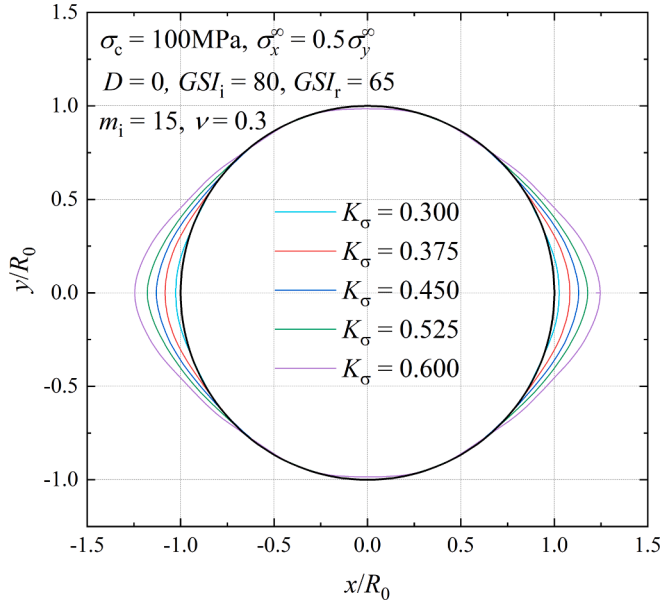


Fig. 14. The plastic zones around a circular tunnel under different values of K_σ .

zone under different joint conditions when $\Delta GSI = GSI_i - GSI_r = 15$. The residual strength GSI_r of the surrounding rock was investigated at values of 65, 60, 55, 50, and 45. And the corresponding plastic zones are depicted in Fig. 13.

It can be observed from Fig. 13 that under identical in-situ stress conditions, if the surrounding rock exhibits good quality (corresponding to a higher GSI_i value), plastic zones generally form in a small region around the tunnel, resembling crescent or U shapes. However, when the quality of the surrounding rock is poor (corresponding to a lower GSI_i value), the plastic zone takes on a V-shape, and the maximum thickness $r_{\max}$ is larger. In contrast to the examples in Section 6.1, it is evident that, regardless of the in-situ stress conditions and GSI values, the plastic zone consistently avoids manifesting a V-shaped configuration. Hence, it is reasonable to conclude that the sudden drop in strength during the failure of surrounding rock is one of the crucial factors contributing to the occurrence of a V-shaped plastic zone.

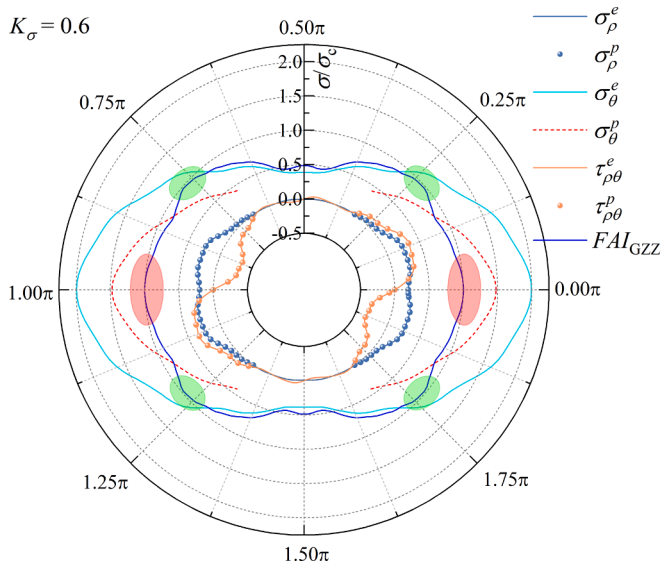


Fig. 15. The in-plane stress components on the inner boundary of the elastic zone and the failure approach index of the surrounding rock.

6.2.2. Influence of K_σ

To further elucidate the evolution laws of the plastic zone shape, we now turn our attention to the variations in the plastic zone under different in-situ stress levels. The parameters are fixed as $GSI_i = 80$, $GSI_r = 65$, $\lambda = 0.5$. The other parameters are the same as those in Section 6.2.1. Consider the plastic zone characteristics for different values of K_σ , namely 0.300, 0.375, 0.450, 0.525, and 0.600. The results are shown in Fig. 14. As shown in Fig. 14, both the shape and size of the plastic zone undergo changes with the increase in K_σ . The plastic zone progressively extends outward, and its boundary approximates a V-shape when the plastic zone is relatively large. Indications suggest a consistent trend in the shape transition of the plastic zone during the development process of a circular tunnel, when contemplating the brittle failure of the surrounding rock. Specifically, the shape evolves from crescent or U-shaped to V-shaped.

The in-plane elastic and plastic stress components on the inner boundary of the elastic zone corresponding to $K_\sigma = 0.6$ in Fig. 14, as well as the failure approach index FAI_{GZZ} at various points on the boundary, are depicted in Fig. 15. These stress components are presented in the orthogonal curvilinear coordinate system described by the conformal mapping function, i.e., Eq. (18). Fig. 15 indicates that the normal and shear stress components on the elastic-plastic interface are both continuous, demonstrating that the stress boundary conditions specified in Eq. (13) are well satisfied. However, there is a significant difference between the elastic tangential stress component σ_θ^e and the plastic tangential stress component σ_θ^p on the elastic-plastic interface. This difference is primarily caused by the brittle failure of the surrounding rock. The solid blue line in Fig. 15 indicates that the FAI_{GZZ} of the surrounding rock outside the elastic-plastic interface varies. In the horizontal direction, FAI_{GZZ} is closest to 1, indicating that this region (highlighted by the red ellipse in Fig. 15) is actually in a high-stress state. Within the range of 0 to 90 degrees, FAI_{GZZ} generally shows a decreasing-increasing-decreasing trend. At an approximate azimuth angle of 45 degrees, there is a specific region (highlighted by the green ellipse in Fig. 15) where the FAI_{GZZ} is close to 1. Actually, the surrounding rock is in a relatively low-stress state in this region.

6.2.3. The maximum thickness of the plastic zone and failure zone

The evolution law of the maximum thickness $r_{\max}$ of the plastic zone under different in-situ stress levels is discussed in this section. The identical computational parameters as outlined in Section 6.2.2 were adopted. The curves depicting the variation concerning the indices K_σ and K_{SCF} , along with the corresponding lines from linear fitting using the

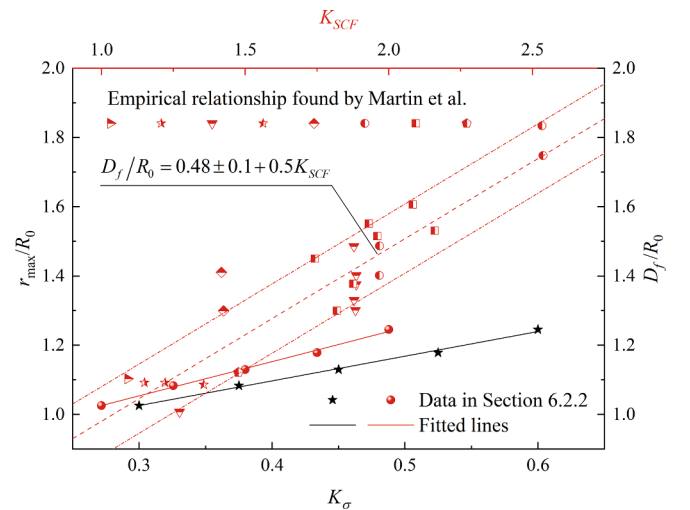


Fig. 16. The field data and the empirical formula for the maximum depth of the failure zone in hard rock provided by Martin et al., and the evolution laws of the plastic zone in hard rock drawn from the data presented in Section 6.2.2.

least squares method, are presented in Fig. 16. In addition, Fig. 16 also provides an extensive dataset based on field tests of the maximum depth D_f of the failure zone in circular tunnels and its relationship with K_{SCF} . Building upon the analysis of this dataset, an empirical relationship has been found by Martin and Christiansson (2009), revealing an approximate linear relationship between D_f / R_0 and K_{SCF} . Emphasis should be placed on the fact that the hard rock parameters and in-situ stress conditions provided in the literature (Martin and Christiansson, 2009) differ from the parameters in this study. Additionally, the literature addresses the depth of the failure zone (or the so-called damage zone) in hard rock tunnels, while this paper specifically investigates the depth of the plastic zone.

Fig. 16 indicates that when assuming the stress–strain relationship of hard rock follows the EBP model, i.e., when considering the brittle strength drop of hard rock, the obtained maximum thickness of the plastic zone still exhibits a noticeable linear correlation with K_σ and K_{SCF} . The depth $r_{\max}$ linearly increases with the increase of K_σ or K_{SCF} . Moreover, an examination of field data from circular hard rock tunnels reveals that the relationship between the maximum depth D_f of failure zone and K_{SCF} can be aptly described by a linear empirical formula. This observation suggests a scientific foundation for estimating the stability of surrounding rock in circular tunnels by considering the magnitudes of indices such as K_σ and K_{SCF} . Moreover, a statistical analysis of extensive in-situ test data indicates that the initial ground stress linearly increases with the growing burial depth H of tunnels (Kang et al., 2010). Hence, it is feasible to establish a linear relationship between $r_{\max}$ and H . Ma et al. (2023) observed that as the H increases, the plastic zone does not uniformly expand in equal proportion, but it approximately increases linearly in a certain direction. Then, they provided a linear fitting formula that can predict the maximum thickness of the plastic zone of circular tunnels at different burial depths.

7. Conclusions

This paper proposes a novel complex variable solution of the plastic zone in deep-buried circular tunnels subjected to non-hydrostatic stress field, considering the 3D strength of the rock mass. Under the premise that the problem discussed is statically determinate, this method is applicable to both the ideal elastic–plastic model and the elastic–brittle–plastic model. This paper considers that the yielding of surrounding rock is caused by compression, that is, there is no tensile failure of surrounding rock. The method can effectively analyze the morphological characteristics of the local plastic zone generated within the surrounding rock, which does not completely encircle the tunnel boundary, and the complete plastic zone that entirely encompasses the tunnel boundary. Through the comparison with existing analytical and semi-analytical methods, this study confirms that the proposed method not only achieves higher computational efficiency but also demonstrates broader applicability. For instance, the new method exhibits unique advantages in predicting surrounding rock stability and plastic zone morphology under non-hydrostatic stress field and varied rock mass parameters. The specific conclusions about circular tunnels are as follows:

Appendix A

Let the term of the series on the right-hand side of Eq. (22) be finite, with both the highest positive and negative power terms set to n . In Eq. (22), $\sigma = e^{i\theta}$ holds true for any $\theta \in [0, 2\pi]$. Thus, within the range of $[0, 2\pi]$, we can select a finite number of θ_j :

$$\theta_j = \frac{2\pi(j-1)}{2n+1}, j = 1, 2, \dots, 2n+1 \quad (\text{A.1})$$

For each θ_j , let the left-hand side of Eq. (22) be:

$$F_j = f(\sigma_j) \quad (\text{A.2})$$

$$\sigma_j = e^{i\theta_j}$$

The failure characteristics of surrounding rock under non-hydrostatic stress. The lateral pressure coefficient λ significantly influences the morphological characteristics of the plastic zone, which consistently reaches its maximum depth in the direction of the smaller principal stress. The plastic zone typically fails to completely encircle the tunnel boundary when λ is small. Importantly, tangential tensile stress occurs in the surrounding rock at the top and bottom of the tunnel if λ is very small, making it susceptible to inducing tensile failure in the surrounding rock. The maximum thickness $r_{\max}$ of the plastic zone monotonically decreases as the difference between the in-situ stresses along the horizontal and vertical directions decreases and reaches its minimum when $\lambda = 1$.

The influence of rock types on the morphological characteristics of the plastic zone. Under the non-hydrostatic stress condition, the plastic zone in soft rock mass typically exhibits a U-shaped configuration, and its shape remains relatively unchanged as K_σ increases. In hard rock tunnels, plastic zones with a small area are generally developed in localized regions. As K_σ increases, the plastic zone undergoes a transformation from a crescent shape to a V-shape, where the brittle drop in the strength of the surrounding rock plays a crucial role in the formation of V-shaped plastic zones. More importantly, it is observed that $r_{\max}$ nonlinearly decreases with the increase of GSI , and there is a significant linear correlation with either K_σ or K_{SCF} . This linear relationship is independent of the relative values of far-field stresses, the in-situ stress level, and the brittleness or ductility characteristics of the surrounding rock. Therefore, K_σ and K_{SCF} can be utilized for a rapid evaluation of the surrounding rock stability in deep-buried circular tunnels.

CRedit authorship contribution statement

Yaocai Ma: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Wuqiang Cai:** Validation, Project administration, Funding acquisition, Conceptualization. **Hehua Zhu:** Supervision, Project administration. **Chenlong Su:** Software, Investigation. **Xiangyang Wei:** Software, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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A system of complex linear equations is then formulated to solve d_0 , d_k , and d_n , we have:

$$\mathbf{ID} = \mathbf{F} \quad (\text{A.3})$$

$$\text{where } \mathbf{I} = \begin{bmatrix} e^{-in\theta_1} & e^{-i(n-1)\theta_1} & \dots & e^{-i\theta_1} & 1 & e^{i\theta_1} & \dots & e^{i(n-1)\theta_1} & e^{in\theta_1} \\ e^{-in\theta_2} & e^{-i(n-1)\theta_2} & \dots & e^{-i\theta_2} & 1 & e^{i\theta_2} & \dots & e^{i(n-1)\theta_2} & e^{in\theta_2} \\ \vdots & \vdots & \dots & \vdots & \vdots & \vdots & \dots & \vdots & \vdots \\ e^{-in\theta_n} & e^{-i(n-1)\theta_n} & \dots & e^{-i\theta_n} & 1 & e^{i\theta_n} & \dots & e^{i(n-1)\theta_n} & e^{in\theta_n} \end{bmatrix},$$

$$\mathbf{D} = [d_n d_{n-1} \dots d_1 d_0 d_1 \dots d_{n-1} d_n]^T, \text{ and } \mathbf{F} = [F_1 F_2 \dots F_n]^T.$$

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